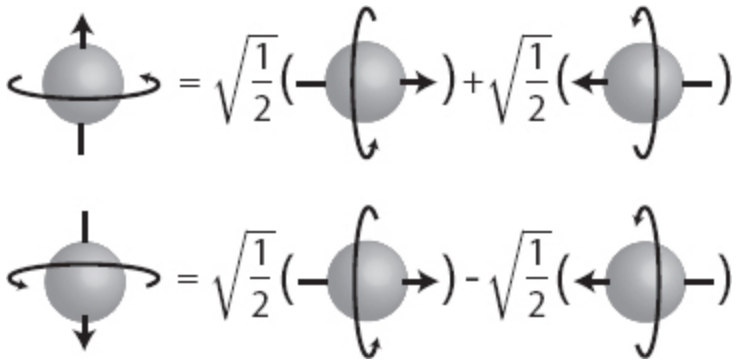


So the idea of being spin-left or spin-right isn’t independent from being spin-up or spin-down; any one possibility can be thought of as a superposition of the others. We say that spin-up and spin-down together form a *basis* for the state of a qubit—any quantum state can be written as a superposition of those two possibilities. But spin-left and spin-right form another basis, distinct but equally good. Writing it one way completely fixes the other way.

Think of this in vector terms. If we draw a two-dimensional plane with spin-up as the horizontal axis and spin-down as the vertical axis, from the above relations we see that spin-right and spin-left point at 45 degrees with respect to them. Given any wave function, we could express it in the up/down basis, but we could equally well express it in the right/left basis. One set of axes is rotated with respect to the other, but they are both perfectly legitimate ways of expressing any vector we like.